

30-DAY DEVOPS LEARNING PLAN

 Daily Time: 2–3 hours

 Goal: Become DevOps-ready with projects

→ WEEK 1: Foundations (Linux + Git + Basics)

“Understand systems before automation”

Day 1 – DevOps Overview

- What is DevOps?
- DevOps lifecycle
- CI vs CD
- DevOps roles & tools
-  Output: Clear DevOps mental model

Day 2 – Linux Basics

- Linux architecture
- File system
- Basic commands (ls, cd, cp, mv, rm, cat)
- Directory structure

Day 3 – Linux Advanced

- Users & groups
- Permissions (chmod, chown)
- Processes (ps, top, kill)
- systemctl, services



Day 4 – Linux Automation

- Shell scripting basics
- Variables, loops, conditions
- Cron jobs
- Log files

Day 5 – Networking Basics

- HTTP/HTTPS
- DNS, IP, ports
- SSH
- Load balancer concept

Day 6 – Git Basics

- git init, clone
- add, commit, push, pull
- GitHub workflow

Day 7 – Git Advanced

- Branching & merging
- Rebase
- Resolving conflicts
- Git best practices

Mini Task:

Push a Linux shell script project to GitHub



→ **WEEK 2: CI/CD + Containers**

"Automate everything"

Day 8 – Build Tools

- npm / Maven / pip
- Build lifecycle
- Artifact concepts

Day 9 – CI/CD Concepts

- Pipelines
- Stages
- Triggers
- Webhooks

Day 10 – Jenkins Basics

- Jenkins setup
- Jobs & pipelines
- Jenkinsfile
- Build automation

Day 11 – CI/CD Advanced

- Environment variables
- Secrets
- Pipeline best practices
- Rollbacks



Day 12 – Docker Basics

- Containers vs VM
- Images & containers
- Docker commands
- Docker Hub

Day 13 – Docker Advanced

- Dockerfile
- Multi-stage builds
- Volumes & networking
- Docker Compose

Day 14 – CI/CD + Docker

- Build Docker image in pipeline
- Push image to registry
- Automated deployment

✓ **Project 1:**

CI/CD pipeline → Dockerized web app



→ **WEEK 3: Kubernetes + Cloud**

“Run apps like production”

Day 15 – Kubernetes Basics

- Kubernetes architecture
- Cluster, nodes, pods
- kubectl basics

Day 16 – Kubernetes Core Objects

- Deployments
- ReplicaSets
- Services
- ConfigMaps & Secrets

Day 17 – Kubernetes Advanced

- Ingress
- Autoscaling
- Resource limits
- Namespaces
- Rolling updates

Day 18 – Helm

- Helm charts
- Values.yaml
- App packaging



Day 19 – Cloud Basics

- Cloud models (IaaS, PaaS, SaaS)
- Regions & availability zones
- IAM concepts

Day 20 – AWS Core Services

- EC2
- S3
- VPC
- Load Balancer
- CloudWatch

Day 21 – Kubernetes on Cloud

- Managed Kubernetes
- Deploy app on cloud cluster
- Expose app publicly

✅ Project 2:

Deploy Docker app on Kubernetes (Cloud)

→ **WEEK 4: IaC, Monitoring, Security & Final Project**

“Think like a real DevOps Engineer”

Day 22 – Infrastructure as Code

- Terraform basics
- Providers & resources
- State files

Day 23 – Terraform Advanced

- Modules
- Remote backend
- Multi-environment setup

Day 24 – Configuration Management

- Ansible basics
- Playbooks
- Inventory
- Roles

Day 25 – Monitoring

- Metrics vs logs
- Prometheus
- Grafana dashboards
- Alerts



Day 26 – Logging

- Centralized logging
- ELK stack basics
- Application logs

Day 27 – DevSecOps

- IAM & RBAC
- Secrets management
- Container scanning
- Code quality tools

Day 28 – Advanced DevOps

- Blue-Green deployments
- Canary releases
- GitOps
- SRE basics

Day 29 – Final Project

End-to-End DevOps Project

- GitHub repo
- CI/CD pipeline
- Docker
- Kubernetes
- Cloud deployment
- Monitoring + security



Day 30 – Interview & Career Prep

- DevOps interview questions
- Resume project descriptions
- Tool revision
- Mock interviews

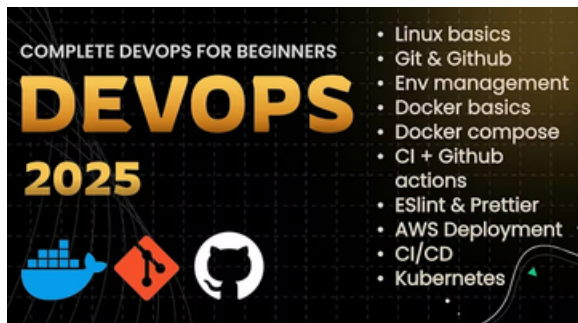
MUST-KNOW TOOLS (After 30 Days)

- Linux • Git • Jenkins
- Docker • Kubernetes
- AWS • Terraform
- Ansible • Prometheus • Grafana

Outcome After 30 Days

- Can build real CI/CD pipelines
- Can deploy apps on Kubernetes
- Can manage cloud infrastructure
- Resume-ready DevOps projects
- Ready for DevOps Intern / Junior DevOps roles

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Git Repo



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